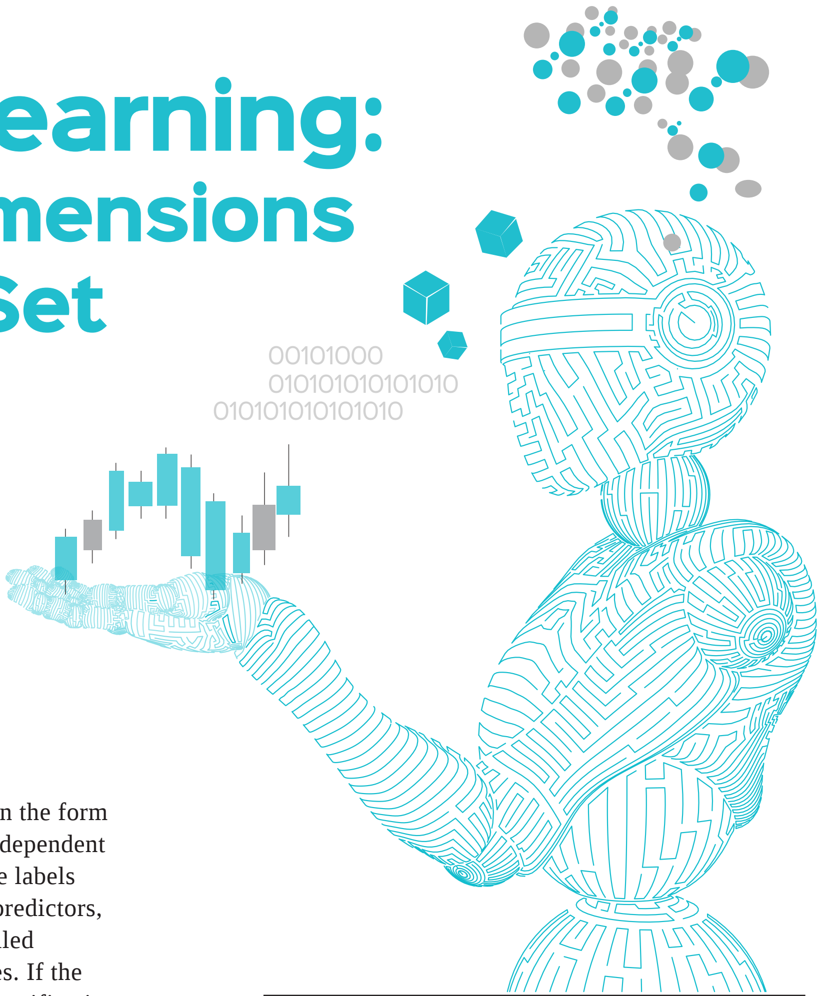
Machine Learning: Reducing Dimensions of the Data Set



Reduction of dimensionality is one of the important processes in machine learning and deep learning. It involves the transformation of input data from high dimensional space to low dimensional space, and retaining meaningful information from the initial input data. Let's look at how this is done.

Data arising out of supervised learning is in the form of attributes and labels. Attributes are independent variables with which we must predict the labels or dependent variables. They are called inputs, predictors, features or independent variables. Labels are called outputs, targets, outcomes or dependent variables. If the labels are categorical, the problem becomes a classification problem, and if they are numerical it becomes a regression problem. The number of input variables is called the dimensionality of the input data.

The need for reduction of dimensionality

The storage space for the input variables is reduced with reduction in dimensionality. This improves the performance of the model. When the number of input variables is more, the computation also increases a lot. In the input data, there may be many features that are correlated while some features may be redundant. The techniques of dimensionality reduction remove the redundant features. They take care of removing the multicollinearity between the features. They also help clustering of the data and make it easier to visualise it in lesser dimensions.

So, apart from speeding up the training time, the reduction of dimensionality helps in data visualisation. Plotting the high dimensional training set on a graph, by reducing it to around two or three dimensions, allows one to gain a few significant insights by data visualisation. This enables observation of patterns and clusters in data.

As the number of dimensions in the data increases, the machine learning problems become more difficult to solve. This particular problem is called curse of dimensionality. As the number of dimensions increase in the training data, the machine learning problem requires a greater number of training examples to generalise the problem. There are many techniques available in machine learning, which allow one to handle the curse of dimensionality and reduce computing costs.

Dimensionality reduction techniques

The various dimensionality reduction techniques available in machine learning can be classified based on how they handle the number of features and the methods they use. Figure 1 gives the different techniques of dimensionality reduction and their classification. These techniques are divided into linear methods and non-linear methods. Linear methods transform the data into linear dimensions. Principle component analysis (PCA), linear discriminant analysis (LDA), non-negative matrix and factorisation are some of the linear methods. Kernel PCA, t-distributed stochastic neighbour embedding (t-SNE), and uniform manifold approximation and projection (UMAP) are some of the non-linear methods. To get a basic understanding of these methods, PCA, LDA, t-SNE and Kernel PCA are briefly described here with an example and a corresponding code snippet.